Finite-Condition Conformal Action Filtering for Shifted Adaptive Cruise Control

Anonymous Authors

Abstract—We study residual-aware conformal action filtering for an adaptive cruise control (ACC) system under a finite set of registered dynamics conditions. The method combines a nominal dynamics model, condition-stratified split-conformal residual quantiles, and a robust action projection with an explicit residual margin. We prove a one-step score-coverage statement under an externally supplied total-variation bound, and a separate conditional one-step safety consequence for a finite registered condition set under an explicit deployment score-law assumption. The paper deliberately does not claim trajectory-level safety, validity for adaptive threshold updates, or robustness outside the registered conditions. The accompanying ACC protocol uses paired randomized resets and separates one-step diagnostics from closed-loop empirical analyses.

Index Terms—Conformal prediction, safe control, adaptive cruise control, distribution shift, robust action filtering.

I. Introduction

Distribution shift can invalidate a nominal predictive model precisely where a controller needs reliable safety margins. Conformal prediction provides finite-sample marginal coverage under exchangeability [1], while shift-aware extensions require explicit assumptions on how the test distribution differs from calibration [2]. Predictive safety filters likewise distinguish a proposed action from the action that can be safely applied [3]. In closed-loop control, it is important to keep those statistical statements distinct from trajectory safety claims.

Conformity-score design can also be adapted to predictive uncertainty models, as in conformalized quantile regression [4]. Here we use a fixed-model next-state residual score because it directly induces the robust one-step margin used by the ACC filter.

We develop RACF as an additive extension of the existing CRSF ACC control path. A nominal model predicts the next physical state, and a conformal quantile bounds its residual score on a calibration population. The action filter then projects a desired action onto robust one-step constraints for a finite, auditable set of ACC dynamics conditions. The residual quantile appears explicitly as a safety margin rather than merely as a reported diagnostic.

This paper has three scoped contributions. First, it states a finite-sample, one-step score-coverage bound under a total-variation assumption on the induced score distributions. Second, it gives a stronger but differently scoped registered-condition result: one calibrated stratum per finite dynamics condition yields a common conservative quantile and a conditional one-step safety corollary. Third, it specifies a paired ACC evaluation that separates static filtering, empirical adaptive updates, and the unfiltered policy. A completed quantile-only update diagnostic produced no adaptive-over-static difference in all eight paired cells, so a moving quantile alone did not improve the registered endpoint. Adaptive updates are empirical and do not inherit the static split-conformal proof. As a separate remedy for the calibration-feedback gap, we also evaluate a pre-frozen, certification-independent envelope with an exact one-step PAC certificate.

II. Residual-Aware Conformal Filtering

Let s, a, and s^+ denote the current state, desired action, and next state. A condition-indexed nominal model $f(s, a, \theta)$ produces a prediction, where θ belongs to a finite registered set Θ. We use the residual score

$$\mathcal{S}_\theta(s, a, s^+) = \|s^+ - f(s, a, \theta)\|_2. \tag{1}$$

For each θ, the calibration scores define the split-conformal order statistic q_θ at target coverage $1-\alpha$. The static filter uses the conservative common threshold $\bar{q} = \max_{\theta\in\Theta} q_\theta$.

Given a desired action a_d, the filter solves a bounded robust projection:

$$\begin{aligned} \tilde{a} \in \arg \min_{a\in[a_{\min},a_{\max}]} \ & \|a - a_d\|_2^2 \\ \text{s.t.} \quad & h(f(s, a, \theta)) \geq L_h \bar{q}, \quad \forall \theta \in \Theta. \end{aligned} \tag{2}$$

Here h is the existing ACC one-step safety margin and L_h is its state-Lipschitz constant used by the safety implication in Section III. If the projection is infeasible, the filter emits the configured bounded emergency action and records the fallback reason. The finite parameter set, action bounds, and fallback behavior are all part of the executable protocol. The one-step safety corollary applies only when the robust projection is feasible; boundedness of the emergency action alone does not give it the same guarantee.

The implementation also supports an adaptive residual buffer. This updates a secondary, empirical threshold after a configured number of transitions. Because the update consumes closed-loop data, it is not covered by the static split-conformal statements below; its role is measured rather than assumed.

III. One-Step Guarantees

Let $Z_i = (s_i, a_i, s_i^+)$ for $i = 1, \ldots, n$ be iid calibration transitions. With a fixed nominal model, write $\mathcal{S}(Z) = \|s^+ - \hat{f}(s, a)\|_2$. Let $\mathcal{S}_{\text{test}}$ be an independent test score

with induced law $P_{\mathcal{S}}^{\text{test}}$ and let $q_\alpha^{(n)}$ be the kth calibration order statistic, where $k = \lceil (n+1)(1-\alpha) \rceil$. If $k > n$, set $q_\alpha^{(n)} = +\infty$.

Theorem 1 (Coverage under bounded score shift). If the calibration scores are iid with induced law $P_{\mathcal{S}}^{\text{cal}}$ and $d_{\text{TV}}(P_{\mathcal{S}}^{\text{cal}}, P_{\mathcal{S}}^{\text{test}}) \le \epsilon$, then

$$\Pr\{\mathcal{S}_{\text{test}} \le q_\alpha^{(n)}\} \ge 1 - \alpha - \epsilon. \tag{3}$$

The probability is jointly over the calibration sample and independent test score. No exchangeability between calibration and shifted test scores is assumed.

Proof. Let $\widetilde{\mathcal{S}}$ be an independent score with law $P_{\mathcal{S}}^{\text{cal}}$. Its exchangeability with the n calibration scores gives $\Pr\{\widetilde{\mathcal{S}} \le q_\alpha^{(n)}\} \ge k/(n+1) \ge 1 - \alpha$ (with the usual random tie-breaking argument). For each realized threshold q, total variation gives $P_{\mathcal{S}}^{\text{test}}((-\infty, q]) \ge P_{\mathcal{S}}^{\text{cal}}((-\infty, q]) - \epsilon$. Taking the expectation over the calibration sample proves the result. □

The required TV bound is an external modeling assumption; MMD and Wasserstein estimates are diagnostics or proxies, not certified TV upper bounds. This theorem does not assert trajectory-level safety. Applying it to a controller whose test action depends on the calibrated threshold additionally requires the stated score-law condition to hold conditional on that calibration sample and resulting policy; that condition is not proved here.

Theorem 2 (Finite registered-condition coverage). For each $\theta \in \Theta$, suppose its calibration and independent test transitions are iid from the same condition-specific law. Then

$$\Pr\{\mathcal{S}_{\theta^\star}(Z^{\theta^\star}_{n_{\theta^\star}+1}) \le \bar{q}\} \ge 1 - \alpha \tag{4}$$

for every fixed registered condition $\theta^\star \in \Theta$.

Proof. Split conformal coverage holds in the $\theta^\star$ stratum at its threshold $q_{\theta^\star}$. Since $\bar{q} \ge q_{\theta^\star}$, the covered event at the stratum threshold is contained in the event at $\bar{q}$. □

The safety implication requires one additional deployment assumption. Fix a calibration realization C, a registered true condition $\theta^\star$, and a deployment context (s, a_d), and let $\tilde{a}$ be a feasible solution of Eq. (2). If

$$\Pr\big\{\|s^+ - f(s, \tilde{a}, \theta^\star)\|_2 \le \bar{q} \mid C, s, a_d, \theta^\star\big\} \ge 1 - \alpha, \tag{5}$$

and h is L_h-Lipschitz with $h(f(s, \tilde{a}, \theta)) \ge L_h \bar{q}$ for every $\theta \in \Theta$, then the next state is safe conditionally with probability at least $1 - \alpha$. This is a conditional one-step corollary, not an automatic consequence for a calibration-dependent closed-loop policy: the marginal finite-condition theorem alone does not establish the displayed conditional score law. Neither result establishes recursive feasibility, simultaneous coverage across conditions, or any trajectory-level guarantee. If the projection is infeasible, the configured emergency action is outside this corollary unless its robust constraint is verified separately.

a) Certification-independent PAC supplement.: The calibration-dependent action map can be decoupled from certification by freezing a deterministic envelope B before collecting deployment-policy transitions. For an iid certification sample of filtered-action scores, an exact one-sided binomial lower confidence bound can then certify the fixed policy's coverage under its stated sampling law. V18 freezes $B = 0.11$ using evidence available before its certification and test streams, and uses a Bonferroni family confidence of 0.95 over the four registered conditions. This is a PAC statement for one fixed policy and sampled one-step contexts; it is not conditional conformal coverage for V16, nor a universal state/action or trajectory guarantee.

Remark 1. In closed-loop deployment, the filtered policy induces the state distribution, so the exchangeability condition used above need not hold. Coverage in Sec. IV is therefore an iid one-step diagnostic. Closed-loop 500-step violation rates are empirical safety measurements, not trajectory-level coverage guarantees.

IV. ACC Evaluation Protocol

The evaluation is an ACC-only partial rollout, not a full benchmark. The registered conditions are nominal (1500 kg, 0° grade), mass shift (1800 kg, 0°), slope shift (1500 kg, 3°), and combined shift (1800 kg, 2°). Each episode has the environment's native 500-step horizon. Static and adaptive conformal filters are compared with the raw frozen policy under shared seeds, randomized initial states, environment parameters, and inference settings.

For V4, each condition has 1000 independently sampled one-step calibration transitions. Two scenarios (brake and cut-in) use 100 paired trials per condition and scenario. Raw, robust-no-margin, static conformal, and adaptive conformal filtering are evaluated under the same reset seed and frozen-policy configuration. The primary endpoint is the paired per-step safety-violation comparison between adaptive and static filtering. Paired bootstrap confidence intervals, one-sided paired sign-flip tests, and Holm correction are applied to the eight primary comparisons. The separate no-margin mechanism ablation uses its own eight-test Holm family; collision, projection, fallback, and per-step latency are secondary endpoints.

A. Coverage Verification (iid one-step diagnostic)

Using the maximum registered-condition calibration threshold $\bar{q} = 0.09079$, a 500-transition held-out diagnostic per condition observed score coverages of 0.896 (nominal), 1.000 (mass shift), 0.894 (slope shift), and 1.000 (combined) at target 0.9. An additional unregistered light-mass (1200 kg) stress diagnostic reused this frozen maximum threshold and observed coverage 0.724. It is deliberately outside the finite registered-condition theorem. Figure 1 also reports residual-score RBF MMD as a shift proxy: light mass has 0.07080 versus 0.00686–0.00956 for the four registered held-out conditions. MMD is not a certified TV

upper bound. These are iid diagnostic partial rollouts, not trajectory-level or adaptive-safety results.

B. Registered-grid fixed-envelope certificate

To address the calibration-dependent deployment gap, V18 selected and froze a deterministic envelope $B = 0.11$ from earlier evidence before its certification and test streams, then ran the seed-0 mixed-IQL policy without reading any certification score. One four-GPU partial rollout generated 1000 independent certification transitions and 500 held-out transitions for each of the four registered conditions, using disjoint reset streams. All certification and held-out coverages were 1.000; the Bonferroni exact-binomial lower bound was 0.99563 per condition at simultaneous family confidence 0.95, above the 0.90 target. Replayed actions matched the stored actions exactly, all actions stayed in bounds, and no fallback occurred. However, the V18 controller grid crossed grades 0° and 3° and therefore omitted the actual combined-condition grade of 2°. Its score coverage result remains a four-condition iid diagnostic, but its finite-grid safety implication does not cover the combined condition. V24 prospectively froze the exact four condition pairs before collecting disjoint streams. In V24, all four certification sets covered 1000/1000 scores and all held-out sets covered 500/500. The simultaneous exact-binomial lower bound was 0.99563 per condition; the recorded grid hash matched, actions replayed exactly, fallback was zero, and the minimum sampled robust margin was 6.773. Thus the sampled finite-grid check now includes the actual combined condition. This is still a fixed-policy iid one-step certificate, not conditional conformal coverage for V16, policy-general efficacy, or trajectory-level safety.

C. Closed-Loop Safety (500-step partial rollout)

The registered V4 partial rollout completed all four conditions and both scenarios, yielding 3200 trial–method records. Adaptive thresholds did update: the final threshold ranged from 0.01997 to 0.03843 across trials, versus the static initialization of 0.09079. Nevertheless, adaptive and static filtering had identical paired per-step violation rates in all eight cells (difference zero, sign-flip and Holm-adjusted $p = 1$). Thus this protocol does not support an adaptive-over-static safety claim.

Relative to the raw frozen policy, static filtering eliminated observed episode-level time-headway violations in nominal and mass-shift braking. Exact paired McNemar tests, Holm-corrected over the 16 raw–filter comparisons, give $p_{\text{Holm}} = 1.59 \times 10^{-12}$ and 6.62×10^{-24}, respectively; these raw–static comparisons are multiplicity-controlled secondary evidence, not the V4 adaptive–static primary family. Cut-in reductions are reported descriptively because the exogenous insertion produces violations before a post-event filter can causally prevent them; they are not presented as episode-level significance claims.

TABLE I
V16 prospectively specified primary-cell results. Counts are violating episodes out of 100 paired ACC brake trials; Δ is raw minus static. The listed p values are exact McNemar tests with the three-cell Holm correction.

Policy	Condition	Raw	Static	Δ (95% CI)	Holm p
P0	Nominal	42/100	0/100	0.42 [0.33, 0.51]	9.09×10^{-13}
P0	Mass shift	75/100	0/100	0.75 [0.66, 0.83]	1.59×10^{-22}
P3	Mass shift	29/100	0/100	0.29 [0.20, 0.38]	3.73×10^{-9}

TABLE II
V4 scoped evidence. Safety rates average the two scenarios; timing is a fixed-state mock test rather than an end-to-end comparison.

Condition	Coverage	Avg. raw violation	Avg. static violation	Avg. adaptive violation
combined	1.000	0.026	0.018	0.018
mass shift	1.000	0.092	0.023	0.023
nominal	0.896	0.065	0.020	0.020
slope shift	0.894	0.022	0.015	0.015
Fixed-state latency (ms; mock test)				static 1.993; adaptive 1.544; mpc 0.280

To assess policy dependence, a separate four-policy ACC brake partial rollout ran all pre-existing mixed-IQL seeds under nominal and mass-shift conditions. The raw safety pressure was heterogeneous: seeds 1–2 had no observed raw violations, whereas seed 0 and seed 3 had pressure cells. Those earlier qualification runs identified the pressure cells; before V16, the fresh reset stream and three-cell family (seed-0 nominal/mass-shift and seed-3 mass-shift) were prospectively specified in project records. Across 100 paired trials per cell, static filtering reduced episodes from 42/100 to 0/100, 75/100 to 0/100, and 29/100 to 0/100, respectively; three-cell Holm-adjusted exact McNemar p-values were 9.09×10^{-13}, 1.59×10^{-22}, and 3.73×10^{-9}. Collisions were 0/100 for both methods in all three cells, so the evidence is for the environment's time-headway safety_violation endpoint, not collision reduction. This prospectively specified ACC brake partial rollout confirms the three named policy-condition cells only; it does not support policy-general efficacy.

The mechanism ablation isolates the conformal residual margin. Relative to the static filter, removing that margin increased mean per-step violation by 0.00518 in nominal braking (95% CI [0.00366, 0.00684]) and 0.01128 in mass-shift braking (95% CI [0.00940, 0.01328]); both have $p_{\text{Holm}} = 8.00 \times 10^{-5}$. The other six mechanism cells are null. This supports a condition- and scenario-qualified effect of the margin, not a universal safety claim.

D. Implementation timing diagnostic

On one current compute node, four Slurm tasks each received one GPU allocation and executed 1000 fixed-state, warmed controller calls under the V4 model and threshold. The mean latency across task-level measurements was 1.993 ms for static filtering, 1.544 ms for adaptive filtering, and 0.280 ms for the existing OSQP MPC controller. This is explicitly a fixed-state mock test: it excludes end-to-end policy inference, selector/window maintenance, and tail latency. It does not establish an MPC comparison or real-time claim.

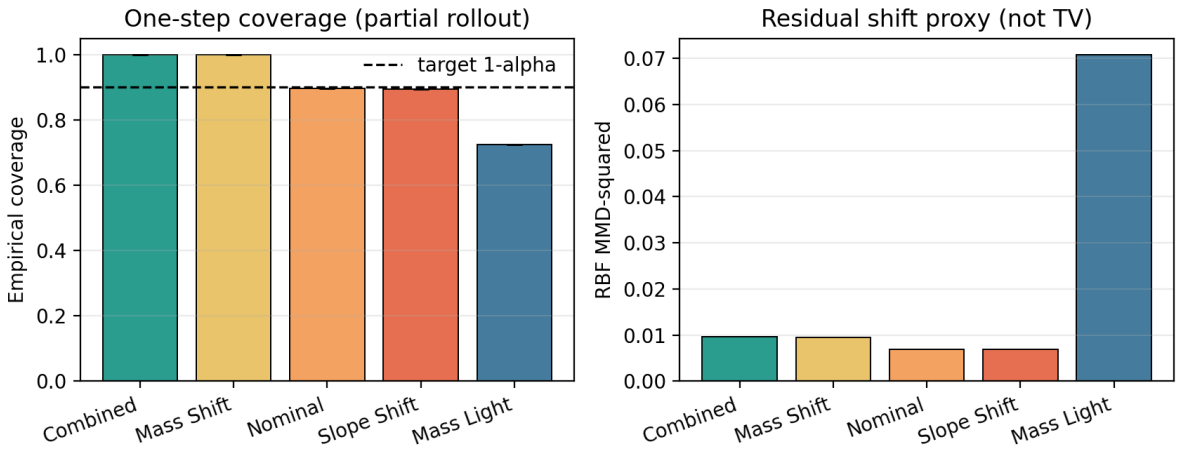


Fig. 1. V5 iid one-step diagnostic under the fixed actuator-gain model. Each condition uses 500 independently sampled held-out transitions. The right panel is RBF MMD-squared from 1000 pooled registered-condition calibration residuals; it is a shift proxy, not a TV bound.

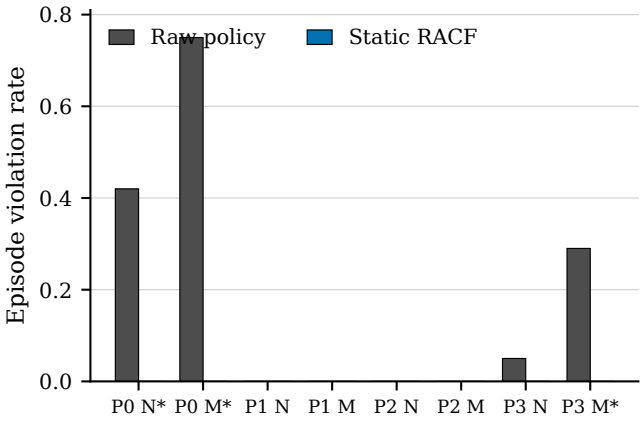


Fig. 2. V16 ACC brake partial-rollout episode-violation rates for all four pre-existing policy seeds. N denotes nominal and M mass shift; stars mark the three prospectively specified primary cells. The zero-pressure cells are retained rather than interpreted as efficacy evidence.

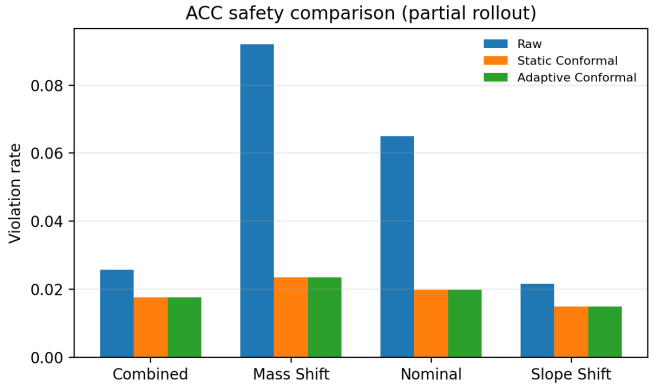


Fig. 3. V4 per-step safety-violation rates averaged over brake and cut-in trials. Static and adaptive are visually coincident in all conditions.

V. Limitations and Reproducibility Boundaries

The bounded-shift theorem depends on exchangeability and an externally valid TV bound on score distributions. The finite-condition result applies only to registered parameter samples with condition-specific calibration data. Neither result covers unregistered dynamics, adaptive updates, recursive feasibility, or trajectory safety. For a deployed action policy that depends on the calibration sample, the required score-law condition must also hold conditional on that sample and policy; this is not established by the current proof. The current evidence is restricted to an ACC partial rollout; it must not be generalized to arbitrary policies, multiple tasks, or a comparison claiming superiority over MPC.

The light-mass one-step diagnostic intentionally lies outside the registered condition set. Its observed coverage and residual MMD are evidence of a stress case, not a coverage theorem extension; RBF MMD is not used as a TV bound.

A separate V17 stratified iid diagnostic used the same registered global threshold and found heterogeneous score coverage across predeclared ego-speed and action strata (lowest observed coverage: 0.800, nominal low-action, $n = 175$, exact 95% CI $[0.733, 0.857]$). This does not establish the conditional deployment score law required by the safety corollary; it records the gap for future work.

V18 addresses that data-flow gap only for a deterministic fixed envelope and the seed-0 mixed-IQL policy. Its PAC lower bound is valid for the stated iid one-step score sampling law and four named data conditions. Its controller grid used grades 0° and 3°, omitting the combined condition's 2° grade, so V18 does not establish the finite-grid safety implication for that condition. Even on listed grid points, robust feasibility is checked on sampled

contexts rather than proved over the continuous ACC state space. It therefore does not extend to calibration-dependent adaptive policies, other policy checkpoints, unregistered conditions, or trajectories. The prospectively frozen V24 protocol replaced that Cartesian grid with the exact four registered mass–grade pairs. Its certificate passed after checking the grid hash, exact action replay, nonnegative sampled robust margins, and zero fallback, but remains limited to the fixed seed-0 policy and sampled iid one-step contexts.

All reported experiment artifacts include a manifest with array shapes, seeds, checksums, protocol version, and data ranges. The V4 condition–scenario family and the separate V16 policy–condition brake family each require their own prespecified 100 paired trials, successful manifest validation, and multiplicity-corrected analysis; neither is a full benchmark.

VI. Conclusion

We presented a finite-condition residual-aware conformal action filter for shifted ACC. The theoretical statements are deliberately one-step and assumption-bound. In the V4 ACC partial rollout, the conformal residual margin reduced the environment's time-headway violation endpoint in seed-0 nominal and mass-shift braking; a later fresh-reset prospectively specified partial rollout confirmed the static effect for seed-3 mass shift, while seeds 1–2 had no raw safety pressure. Adaptive threshold updates did not improve safety over the static filter in the eight registered condition–scenario cells. Fixed-state timing characterizes implementation overhead without establishing a real-time or MPC runtime comparison. Extending the V24 exact-grid fixed-envelope certificate to additional policies and state/action support, as well as extending the evaluation to a broader task suite and end-to-end hardware benchmark, remains future work.

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